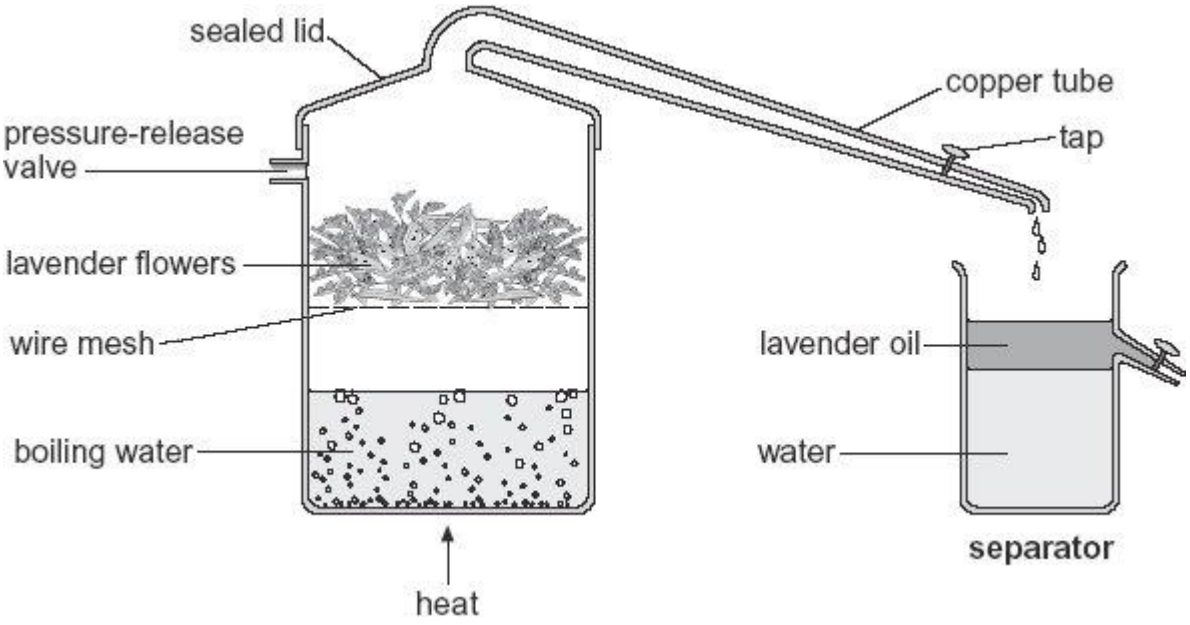
Q1.

Lavender oil is a perfume obtained from lavender flowers.
Steam at 100°C is passed through the flowers in the apparatus below.



not to scale

Water vapour and lavender oil vapour pass down a copper tube towards a separator.

(a) (i) The lavender flowers are heated in a container with a sealed lid.

Why must the lid be sealed?

.....

.....

1 mark

(ii) What would happen if the container did **not** have a pressure-release valve?

.....

.....

1 mark

(b) Lavender oil vapour and water vapour cool as they pass down the copper tube.
A mixture of lavender oil and water collects in the separator.

(i) What is the change in the physical state of both lavender oil vapour and water vapour as they cool?

from ..... to .....

1 mark

(ii) Look at the separator.

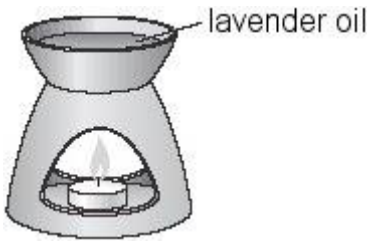
How does this show that the water is denser than lavender oil?

.....

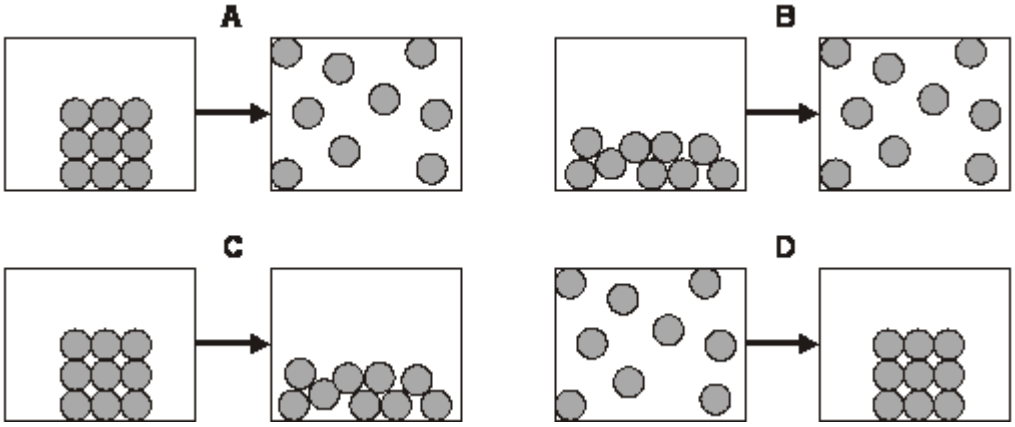
.....

1 mark

(c) Rosie poured some lavender oil into an oil burner.
She heated it with a candle.



The oil changed state.



Which diagram represents this change of state?
Write the letter.

.....

1 mark
maximum 5 marks

Q1.

(a) (i) any **one** from

- to stop water vapour **or** steam escaping
  *accept 'gas **or** vapour **or** particles could escape'*
  *'to stop water escaping **or** spilling out' is insufficient*

- to stop lavender oil vapour escaping
  accept 'to stop oil escaping'
  accept 'to make sure the oil goes down the tube'
  *accept 'the oil **or** vapours could not be collected'*
  'to stop the smell escaping' is insufficient

- to reduce heat loss
  *accept 'to stop heat loss' **or** 'to keep heat in'*
  accept 'to stop the heat escaping'
  accept 'to maintain the pressure'

1 (L5)

(ii) any **one** from

- it could explode

- it could break
  accept 'the lid might come off'
  accept 'so the pressure does not break the seal'
  'the pressure would be too high' is insufficient
  'it would get too hot' is insufficient

1 (L5)

(b) (i) • *from* gas *to* liquid
  ***both** answers are required for the mark*
  answers must be in the correct order

1 (L5)

(ii) any **one** from

- it sinks
  accept 'lavender oil floats'

- it is at the bottom
  accept 'the oil is on the top'
  'they are not mixed' is insufficient

1 (L6)

(c) • B
  if more than one letter is identified, award no mark

1 (L6)

[5]